

Basic Principles of Medicine 2

Module: Nervous System & Behavioral Science | DLA 7

Pharyngeal Apparatus: Arches, Grooves, & Pouches

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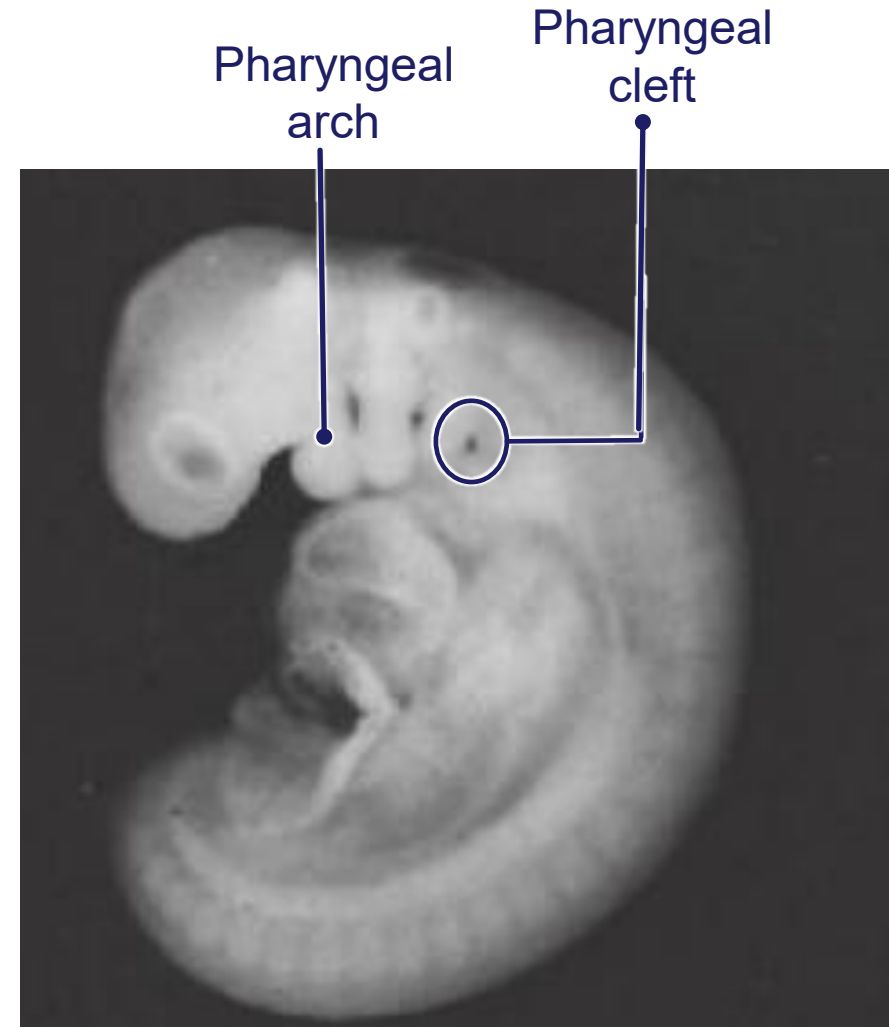
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Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0007	Describe the formation of the pharyngeal arches, grooves, and pouches
SOM.MKII.BPM2.1.NB.1.ANAT.0008	Describe the components of the pharyngeal apparatus
SOM.MKII.BPM2.1.NB.1.ANAT.0009	Describe the structures derived from each pharyngeal arch, pouch and groove
SOM.MKII.BPM2.1.NB.1.ANAT.0010	Describe the cartilage, nerve, and blood vessel of each pharyngeal arch
SOM.MKII.BPM2.1.NB.1.ANAT.0011	Describe the development of the stomodeum

Embryonic Pharynx

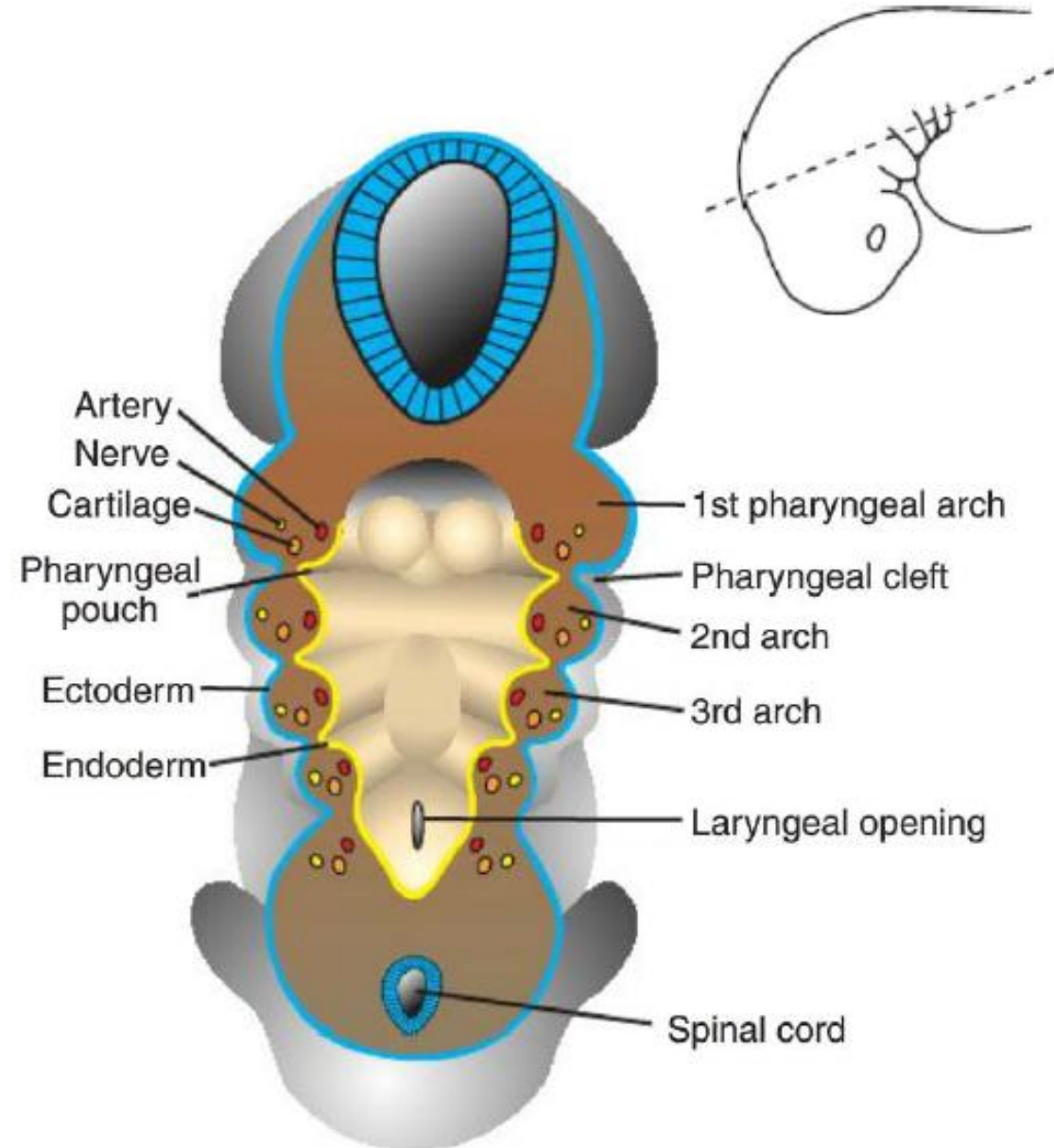
- **Neural crest cells** migrate to meet the most cranial portion of the foregut
- Form elevations and ridges of mesenchyme called **pharyngeal arches**
 - Create scaffolding at the lateral walls of the primordial pharynx
 - Visible in the 4th week of development (around day 26)
- 6 arches formed in total
 - 2 are rudimentary



Pharyngeal (Branchial) Apparatus

Consists of

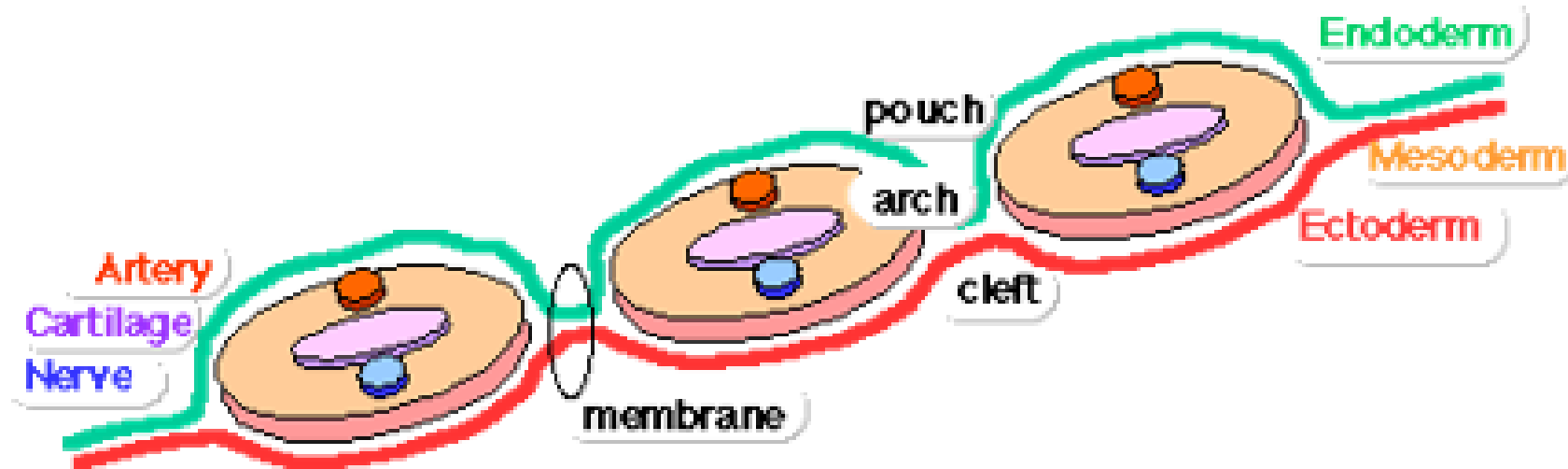
- **Pharyngeal arches:** elevations and ridges
 - Contain **core of mesenchyme**
- **Pharyngeal pouches:** clefts on inner wall of pharynx
 - Lined by **endoderm**
- **Pharyngeal grooves:** clefts that separate arches externally
 - Lined by **ectoderm**
- **Pharyngeal membranes:** where *ectoderm and endoderm meet* between arches



Pharyngeal Arch Content

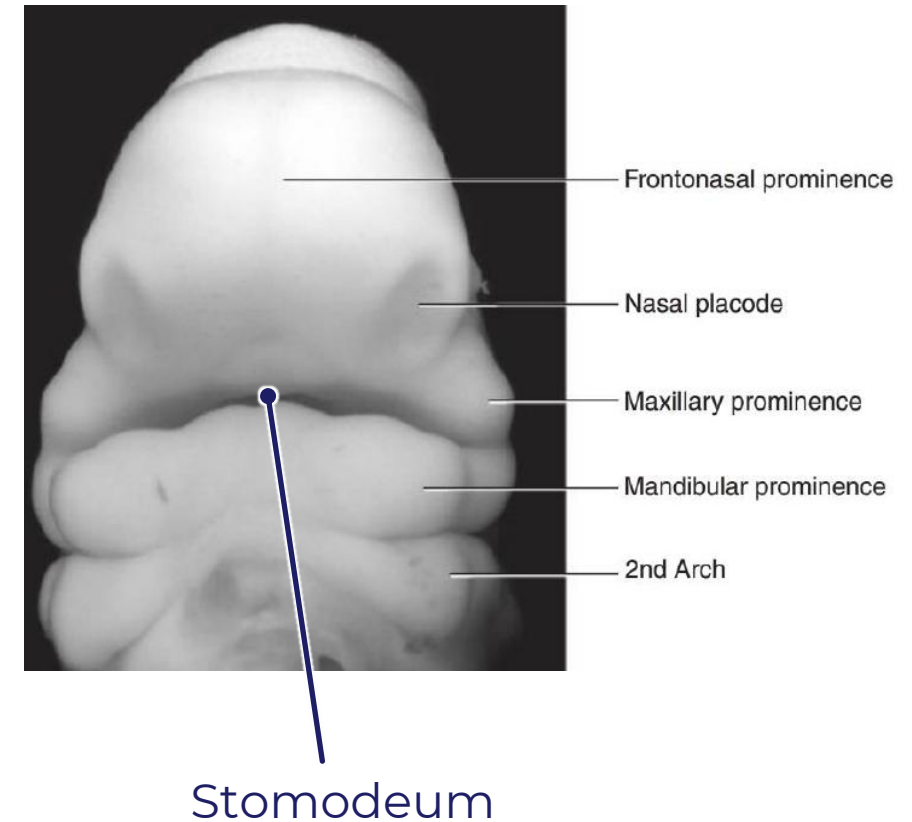
A typical arch has the following components:

- **Aortic arch artery** arising from the truncus arteriosus that courses around the primordial pharynx
- **Cartilaginous rod** that forms the skeleton of the arch
- **Muscular precursors** to head and neck muscles
- **A nerve** supplying the mucosa and muscles of the arch



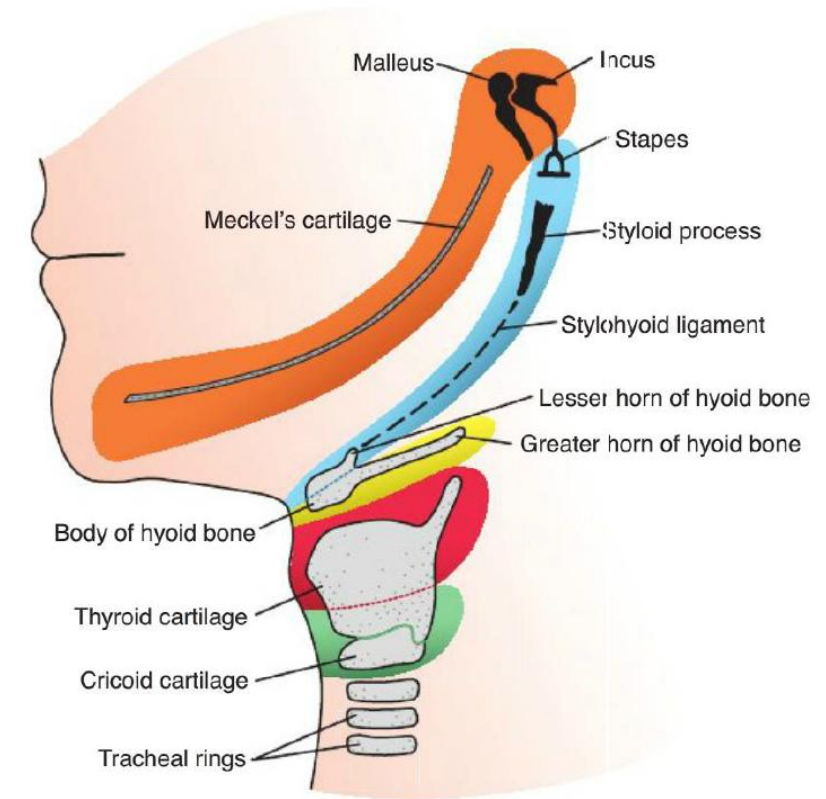
Formation of stomodeum

- Center of the face is formed by the stomodeum, which is surrounded by the **first pair** of pharyngeal arches
- 1st pharyngeal arch **splits into two** bilaterally
 - Maxillary prominence
 - Mandibular prominence – fuse across the midline to form the lower jaw
- Stomodeum: primordium of the **mouth**
 - Initially appears as a depression of surface ectoderm
 - Separated from foregut by **oropharyngeal membrane**
 - Ruptures to allow primordial pharynx and foregut to communicate



Pharyngeal Arch - Cartilages

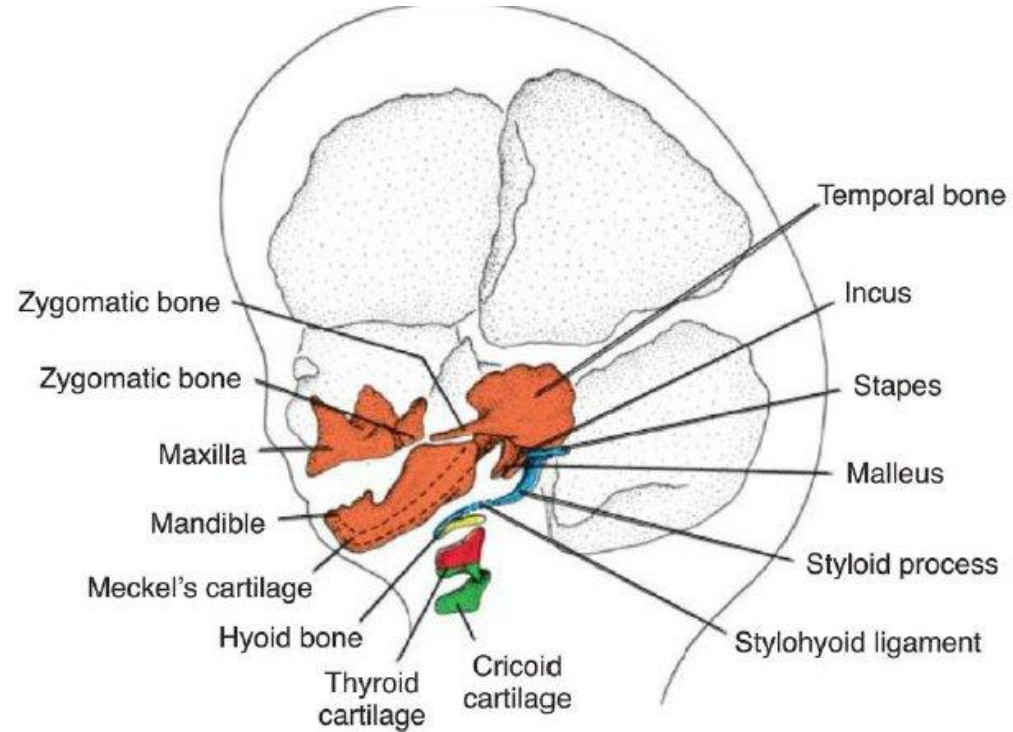
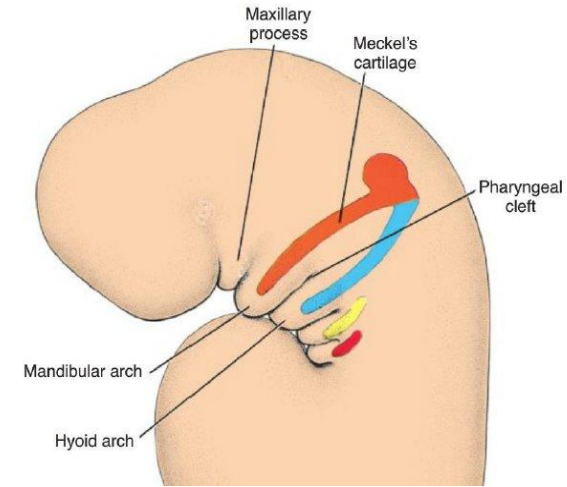
- Meckel's cartilage (from 1st arch)
 - provides the framework for development of the mandible
 - degenerates and forms the mandibular canal and sphenomandibular ligament
- 2nd arch cartilage
 - Contributes to stapes, styloid process, stylohyoid ligament, lesser cornu of hyoid bone
- 3rd arch cartilage
 - Ossifies to form greater cornu of hyoid bone
- 4th and 6th arch cartilage
 - fuse to form laryngeal cartilages (except epiglottis)



1st Pharyngeal Arch bones

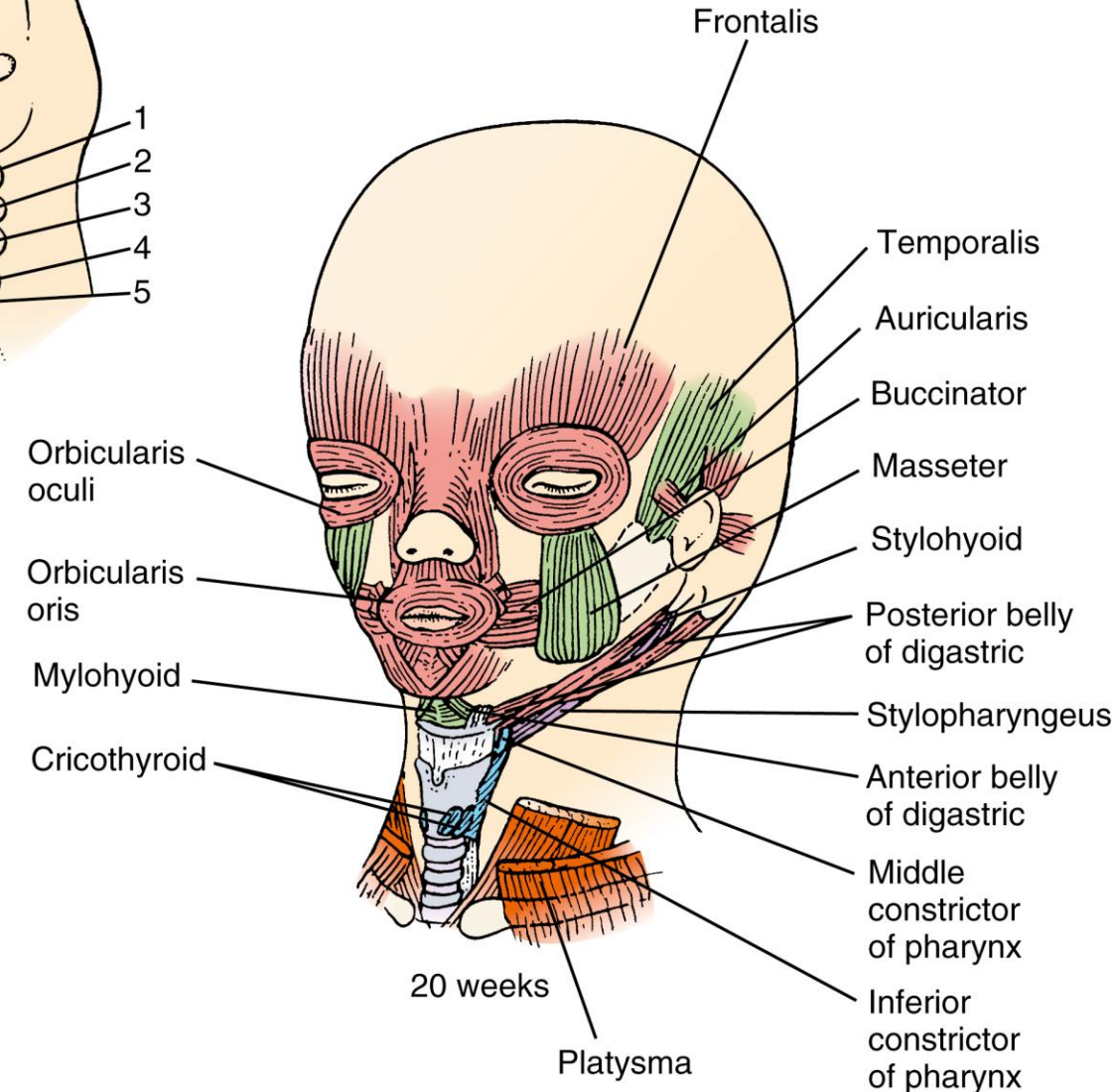
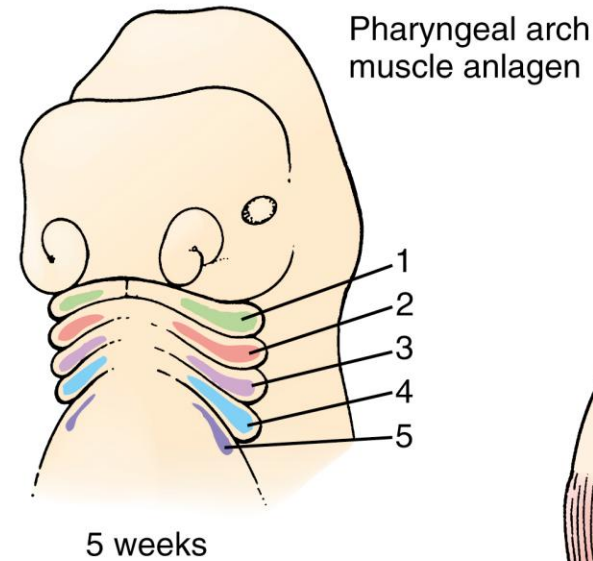
Divides into maxillary and mandibular prominences

- Maxillary prominence
 - *maxilla*
 - *Zygomatic bone*
 - portion of vomer
- Mandibular prominence
 - *mandible*
 - *squamous temporal bone*
 - sphenomandibular ligament
 - *part of middle ear*
 - part of external ear



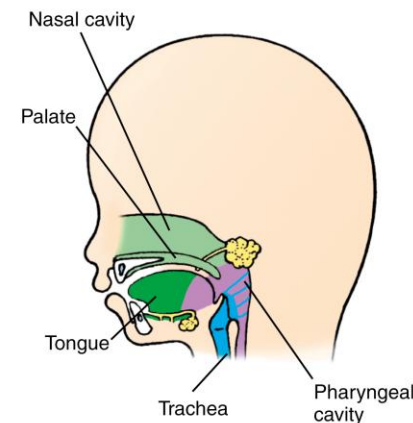
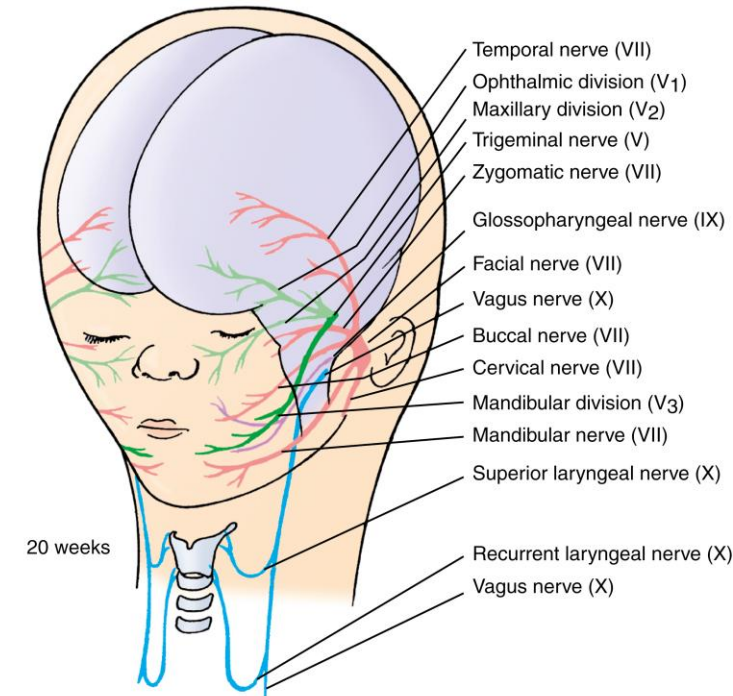
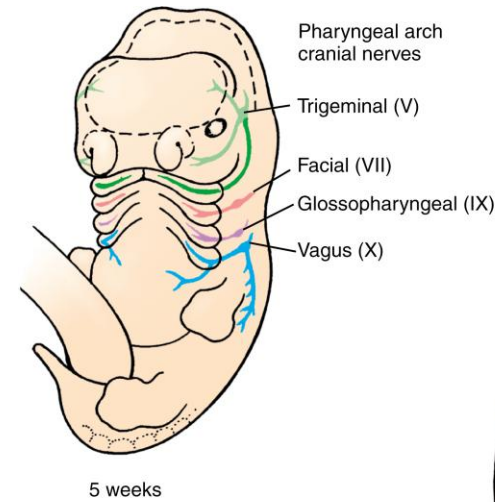
Pharyngeal Arch - Muscles

- **1st arch** forms
 - tensor tympani
 - tensor veli palatini
 - mylohyoid
 - muscles of mastication
 - anterior belly of digastric
- **2nd arch** forms
 - posterior belly of digastric
 - stapedius
 - stylohyoid
 - muscles of facial expression
- **3rd arch** forms stylopharyngeus muscles
- **4th and 6th arches** combined form pharynx and larynx muscles



Pharyngeal Arch - Nerves

- **Special visceral efferent** components
 - supply branchial-derived muscles
- **Special visceral afferent** components
 - supply dermis and and mucus membranes of head and neck
- Caudal branches of trigeminal nerve (**V2** and **V3**)
 - sensory supply to the face, nasal cavity, hard palate, teeth and tongue
- Nerves of 2nd – 6th arches
 - supply the mucus membranes of tongue, pharynx and larynx
- **Superior laryngeal (X)** supplies the 4th arch
- **Recurrent laryngeal (X)** supplies the 6th arch



Sensory innervation of nasal and oral cavities and pharynx

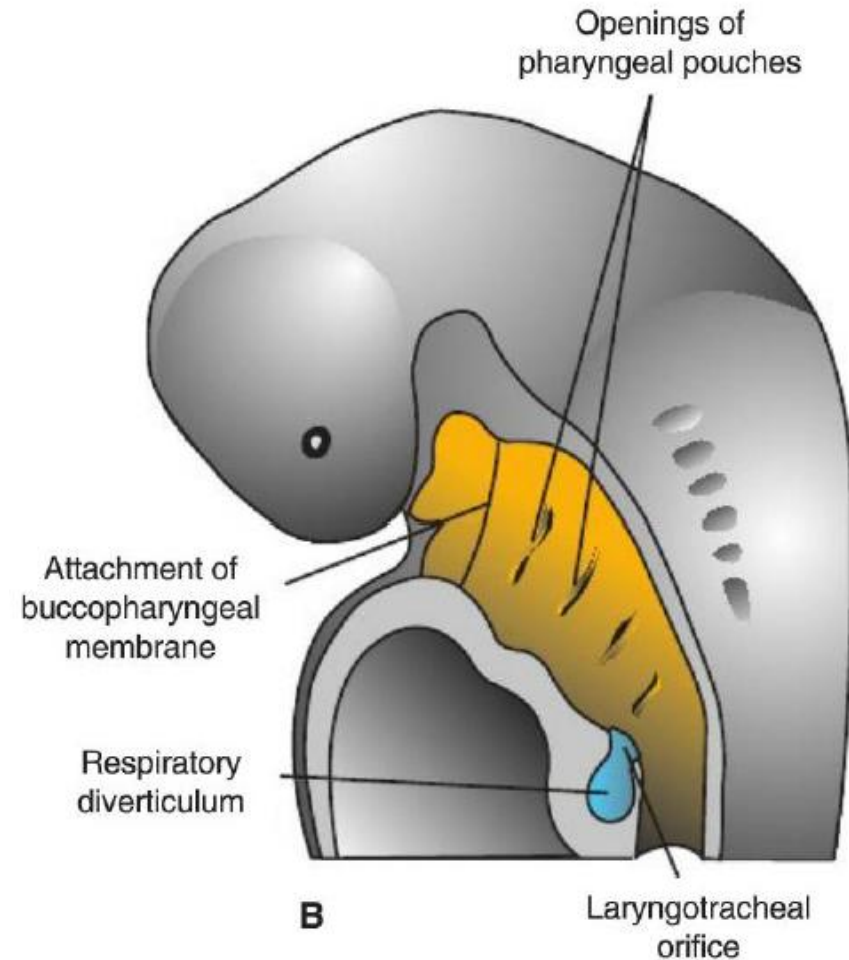
- Maxillary division of trigeminal (V2)
- Mandibular division of trigeminal (V3) (plus chorda tympani from facial nerve)
- Glossopharyngeal (IX)
- Vagus (X)

Arch - Derivatives

Arch	Cranial nerve	Artery	Skeleton	Muscles
1st	Trigeminal (V2, V3)	Maxillary	Malleus, incus, spheno-mandibular ligament, maxilla, mandible, squamous temporal bone, zygomatic bone, portion of vomer	Mastication, tensor veli palatini, tensor tympani, anterior digastric, mylohyoid
2nd	Facial (V)		Stapes, styloid process, stylohyoid ligament, hyoid (lesser horn and upper body)	Stapedius, stylohyoid, posterior digastric
3rd	Glosso-pharyngeal (IX)	Common and internal carotid	Hyoid (greater horn and lower body)	Stylopharyngeus
4th	Vagus (X) - superior laryngeal	Aortic arch, right subclavian	Laryngeal cartilages	Pharyngeal constrictors, levator veli palatini, cricothyroid
6th	Vagus (X) - recurrent laryngeal	Ductus arteriosus, pulmonary arteries	Laryngeal cartilages	Intrinsic muscles of larynx, striated muscles of esophagus

Pharyngeal Pouches

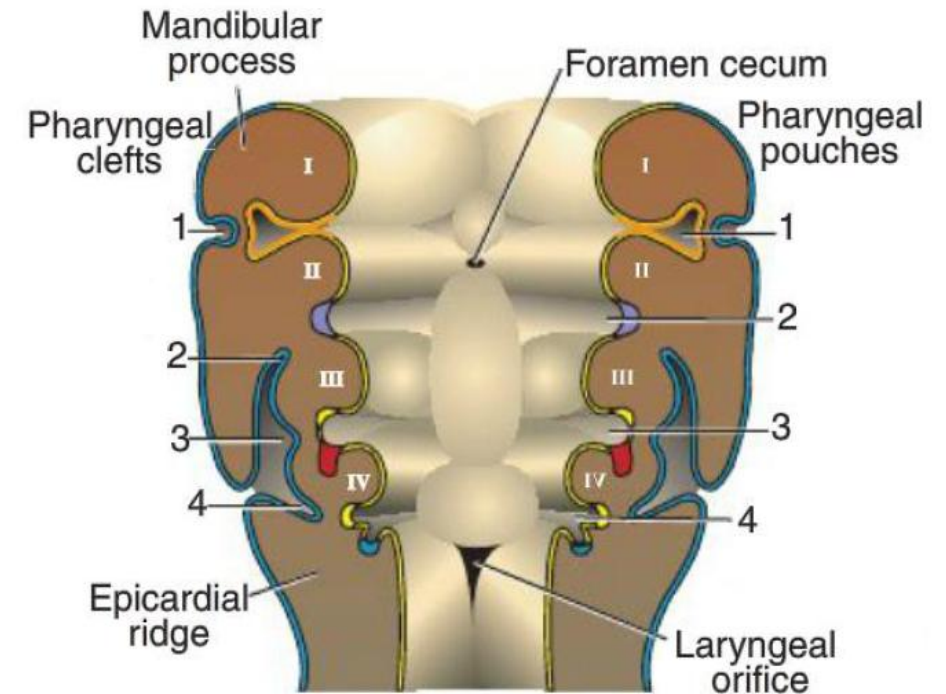
- **1st pharyngeal pouch**
 - contributes to tympanic membrane, tympanic cavity, and mastoid antrum
- **2nd pharyngeal pouch**
 - mostly obliterated as palatine tonsil develops
 - remains form tonsillar sinus and tonsillar crypts
- **3rd pharyngeal pouch**
 - ventrally forms thymus
 - dorsally differentiates into inferior parathyroid gland
- **4th pharyngeal pouch**
 - dorsal epithelium forms superior parathyroid
 - ventral region gives rise to *ultimobranchial body*
 - incorporated into thyroid gland



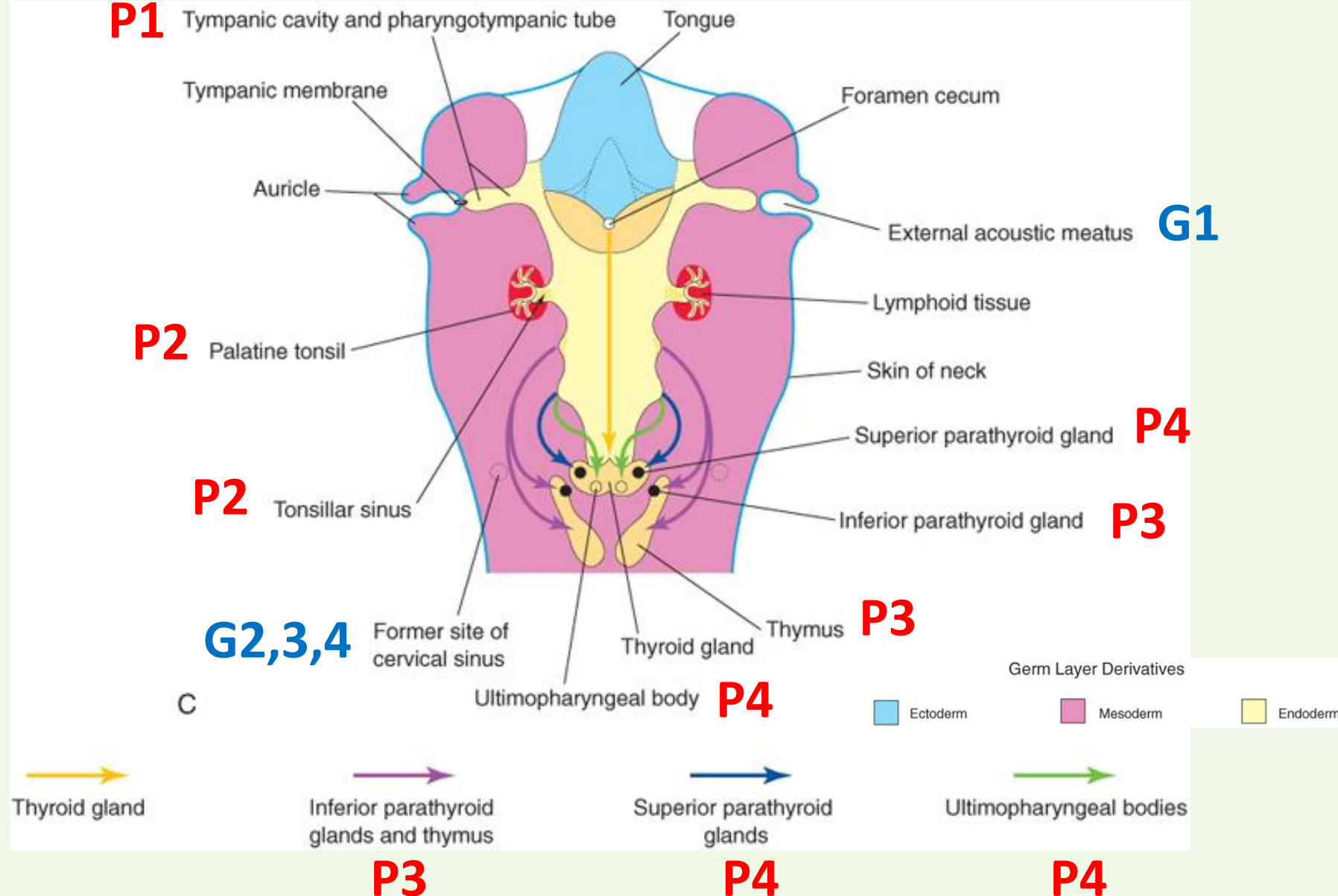
Pharyngeal Grooves (Clefts)

4 pharyngeal clefts at 5 weeks

- **2nd arch** mesenchymal tissue actively proliferates
 - begins to overlap 3rd and 4th arches until merges with epicardial ridge in lower neck
 - **2nd, 3rd, and 4th clefts** lose contact with outside
- **1st cleft** dorsal part penetrates underlying mesenchyme
 - gives rise to external acoustic meatus



Pharyngeal pouches and grooves - summary



Grooves and Pouches - Derivatives

Grooves	
1st	External acoustic meatus
2nd	Obliterated by proliferation of 2nd arch to join the 6th arch
3rd	Obliterated by proliferation of 2nd arch to join the 6th arch
4th	Obliterated by proliferation of 2nd arch to join the 6th arch

Pouches	
1st	Tubotympanic recess, middle ear cavity, mastoid antrum
2nd	Palatine tonsils, tonsillar fossa
3rd	Thymus, inferior parathyroid glands
4th	Superior parathyroid glands, ultimopharyngeal body (parafollicular cells of thyroid)